

Lecture - 5

② Solve the system of equations

$$2x + 3y = 8$$

$$4x + 6y = 16$$

The augmented matrix form of the above system is given by

$$\left(\begin{array}{cc|c} 2 & 3 & 8 \\ 4 & 6 & 16 \end{array} \right)$$

Row operations

$$\left(\begin{array}{cc|c} 2 & 3 & 8 \\ 4 & 6 & 16 \end{array} \right) \xrightarrow{R_1 \rightarrow \frac{1}{2}R_1} \left(\begin{array}{cc|c} 1 & \frac{3}{2} & 4 \\ 4 & 6 & 16 \end{array} \right)$$

$$\left(\begin{array}{cc|c} 1 & \frac{3}{2} & 4 \\ 4 & 6 & 16 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - 4R_1} \left(\begin{array}{cc|c} 1 & \frac{3}{2} & 4 \\ 0 & 0 & 0 \end{array} \right)$$

$$\begin{array}{l} 2x + 3y = 8 \\ 4x + 6y = 16 \end{array}$$

$$\downarrow$$
$$\begin{array}{l} x + \frac{3}{2}y = 4 \\ 4x + 6y = 16 \end{array}$$

mul first
eqn by -4
and add
to second eqn

$$\begin{array}{l} 2x + 3y = 8 \leftarrow \\ 0x + 0y = 0 \end{array}$$

Solution is given by $\left\{ \left(t, \frac{8-2t}{3} \right) : t \text{ is any number} \right\}$

③ Solve the system of equations

$$2x + 3y = 8$$

$$4x + 6y = 1$$

The augmented matrix form of the above system is given by

$$\left(\begin{array}{cc|c} 2 & 3 & 8 \\ 4 & 6 & 1 \end{array} \right)$$

Row operations

$$\left(\begin{array}{cc|c} 2 & 3 & 8 \\ 4 & 6 & 1 \end{array} \right) \xrightarrow{R_1 \rightarrow \frac{1}{2}R_1} \left(\begin{array}{cc|c} 1 & \frac{3}{2} & 4 \\ 4 & 6 & 1 \end{array} \right)$$

$$\left(\begin{array}{cc|c} 1 & \frac{3}{2} & 4 \\ 4 & 6 & 1 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - 4R_1} \left(\begin{array}{cc|c} 1 & \frac{3}{2} & 4 \\ 0 & 0 & -15 \end{array} \right)$$

$$\begin{array}{l} 2x + 3y = 8 \\ 4x + 6y = 1 \end{array}$$

$$\downarrow$$
$$\begin{array}{l} x + \frac{3}{2}y = 4 \\ 4x + 6y = 1 \end{array}$$

multiply first eqn by -4 and add to the second eqn

$$\boxed{\begin{array}{l} x + \frac{3}{2}y = 4 \\ 0x + 0y = -15 \end{array}}$$

Thus the system has no solutions!

④

Solve the system of equations

$$2x + 5y + 2z = 0$$

$$3x + 2y + z = 2$$

$$4x - y + 3z = 1$$

The augmented matrix is given by

$$\left(\begin{array}{ccc|c} 2 & 5 & 2 & 0 \\ 3 & 2 & 1 & 2 \\ 4 & -1 & 3 & 1 \end{array} \right)$$

Row operations

$$\left(\begin{array}{ccc|c} 2 & 5 & 2 & 0 \\ 3 & 2 & 1 & 2 \\ 4 & -1 & 3 & 1 \end{array} \right) \xrightarrow{R_1 \rightarrow \frac{1}{2} R_1} \left(\begin{array}{ccc|c} 1 & \frac{5}{2} & 1 & 0 \\ 3 & 2 & 1 & 2 \\ 4 & -1 & 3 & 1 \end{array} \right)$$

$$\left(\begin{array}{ccc|c} 1 & \frac{5}{2} & 1 & 0 \\ 3 & 2 & 1 & 2 \\ 4 & -1 & 3 & 1 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - 3R_1} \left(\begin{array}{ccc|c} 1 & \frac{5}{2} & 1 & 0 \\ 0 & -\frac{11}{2} & -2 & 2 \\ 4 & -1 & 3 & 1 \end{array} \right)$$

$$\left(\begin{array}{ccc|c} 1 & \frac{5}{2} & 1 & 0 \\ 0 & -\frac{11}{2} & -2 & 2 \\ 4 & -1 & 3 & 1 \end{array} \right) \xrightarrow{R_3 \rightarrow R_3 - 4R_1} \left(\begin{array}{ccc|c} 1 & \frac{5}{2} & 1 & 0 \\ 0 & -\frac{11}{2} & -2 & 2 \\ 0 & -11 & -1 & 1 \end{array} \right)$$

$$\begin{pmatrix} 1 & \frac{1}{2} & 1 & | & 0 \\ 0 & \frac{1}{2} & -2 & | & 2 \\ 0 & -1 & -1 & | & 1 \end{pmatrix} \xrightarrow{R_2 \rightarrow \frac{-2}{1} R_2} \begin{pmatrix} 1 & \frac{1}{2} & 1 & | & 0 \\ 0 & 1 & \frac{4}{1} & | & \frac{-4}{1} \\ 0 & -1 & -1 & | & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & \frac{1}{2} & 1 & | & 0 \\ 0 & 1 & \frac{4}{1} & | & \frac{-4}{1} \\ 0 & -1 & -1 & | & 1 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 + 1R_2} \begin{pmatrix} 1 & \frac{1}{2} & 1 & | & 0 \\ 0 & 1 & \frac{4}{1} & | & \frac{-4}{1} \\ 0 & 0 & 3 & | & -3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & \frac{1}{2} & 1 & | & 0 \\ 0 & 1 & \frac{4}{1} & | & \frac{-4}{1} \\ 0 & 0 & 3 & | & -3 \end{pmatrix} \xrightarrow{R_3 \rightarrow \frac{1}{3} R_3} \begin{pmatrix} 1 & \frac{1}{2} & 1 & | & 0 \\ 0 & 1 & \frac{4}{1} & | & \frac{-4}{1} \\ 0 & 0 & 1 & | & -1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & \frac{1}{2} & 1 & | & 0 \\ 0 & 1 & \frac{4}{1} & | & \frac{-4}{1} \\ 0 & 0 & 1 & | & -1 \end{pmatrix} \xrightarrow{R_1 \rightarrow R_1 - \frac{5}{2} R_2} \begin{pmatrix} 1 & 0 & \frac{1}{1} & | & \frac{10}{1} \\ 0 & 1 & \frac{4}{1} & | & \frac{-4}{1} \\ 0 & 0 & 1 & | & -1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & \frac{1}{1} & | & \frac{10}{1} \\ 0 & 1 & \frac{4}{1} & | & \frac{-4}{1} \\ 0 & 0 & 1 & | & -1 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 - \frac{4}{1} R_3} \begin{pmatrix} 1 & 0 & \frac{1}{1} & | & \frac{10}{1} \\ 0 & 1 & 0 & | & 0 \\ 0 & 0 & 1 & | & -1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & \frac{1}{1} & | & \frac{10}{1} \\ 0 & 1 & 0 & | & 0 \\ 0 & 0 & 1 & | & -1 \end{pmatrix} \xrightarrow{R_1 \rightarrow R_1 - \frac{1}{1} R_3} \begin{pmatrix} 1 & 0 & 0 & | & 1 \\ 0 & 1 & 0 & | & 0 \\ 0 & 0 & 1 & | & -1 \end{pmatrix}$$

Thus, there is a unique solution for the given system of equations, namely, $\boxed{x=1; y=0; z=-1.}$

5

Solve the system of equations

$$2x + 5y + 2z = 0$$

$$-11y - z = 1$$

$$4x - y + 3z = 1$$

The augmented matrix is given by

$$\left(\begin{array}{ccc|c} 2 & 5 & 2 & 0 \\ 0 & -11 & -1 & 1 \\ 4 & -1 & 3 & 1 \end{array} \right)$$

Row operations

$$\left(\begin{array}{ccc|c} 2 & 5 & 2 & 0 \\ 0 & -11 & -1 & 1 \\ 4 & -1 & 3 & 1 \end{array} \right) \xrightarrow{R_1 \rightarrow \frac{1}{2}R_1} \left(\begin{array}{ccc|c} 1 & \frac{5}{2} & 1 & 0 \\ 0 & -11 & -1 & 1 \\ 4 & -1 & 3 & 1 \end{array} \right)$$

$$\left(\begin{array}{ccc|c} 1 & \frac{5}{2} & 1 & 0 \\ 0 & -11 & -1 & 1 \\ 4 & -1 & 3 & 1 \end{array} \right) \xrightarrow{R_3 \rightarrow R_3 + -4R_1} \left(\begin{array}{ccc|c} 1 & \frac{5}{2} & 1 & 0 \\ 0 & -11 & -1 & 1 \\ 0 & -11 & -1 & 1 \end{array} \right)$$

$$\begin{array}{l} R_2 \rightarrow \frac{-1}{11}R_2 \\ R_3 \rightarrow \frac{-1}{11}R_3 \end{array} \left(\begin{array}{ccc|c} 1 & \frac{5}{2} & 1 & 0 \\ 0 & -11 & -1 & 1 \\ 0 & -11 & -1 & 1 \end{array} \right) \xrightarrow{\quad} \left(\begin{array}{ccc|c} 1 & \frac{5}{2} & 1 & 0 \\ 0 & 1 & \frac{1}{11} & \frac{-1}{11} \\ 0 & 1 & \frac{1}{11} & \frac{-1}{11} \end{array} \right)$$

$$\left(\begin{array}{ccc|c} 1 & \frac{5}{2} & 1 & 0 \\ 0 & 1 & \frac{1}{11} & \frac{-1}{11} \\ 0 & 1 & \frac{1}{11} & \frac{-1}{11} \end{array} \right) \xrightarrow{R_3 \rightarrow R_3 + -R_2} \left(\begin{array}{ccc|c} 1 & \frac{5}{2} & 1 & 0 \\ 0 & 1 & \frac{1}{11} & \frac{-1}{11} \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$\left(\begin{array}{ccc|c} 1 & \frac{5}{2} & 1 & 0 \\ 0 & 1 & \frac{1}{11} & \frac{-1}{11} \\ 0 & 0 & 0 & 0 \end{array} \right) \xrightarrow{R_1 \rightarrow R_1 + \frac{-5}{2} R_2} \left(\begin{array}{ccc|c} 1 & 0 & \frac{17}{22} & \frac{5}{22} \\ 0 & 1 & \frac{1}{11} & \frac{-1}{11} \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Thus we have reduced our system to

$$x + \frac{17}{22} z = \frac{5}{22}$$

$$y + \frac{1}{11} z = \frac{-1}{11}$$

* No further reduction
is possible *

Set $z := t$ and obtain

$$x = \frac{5}{22} - \frac{17}{22}t, \quad y = \frac{-1}{11} - \frac{1}{11}t, \quad z = t$$

Solution set

$$\left\{ \left(\frac{5}{22} - \frac{17}{22}t, \frac{-1}{11} - \frac{1}{11}t, t \right) : t \text{ is any real number} \right\}$$